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Unit 4 : Session 10 : SLO 2 | Pipelining Performance Fill in the Blanks

I. Fill in the Blanks

- throughput is the primary pipelining metric, measured in instructions per cycle (IPC).
- Pipeline stall is the pipeline pauses when dependencies cannot be resolved in time.
- depth (number of stages) is ideal speedup equals the number of pipeline stages under perfect conditions.
- Structural is hazards where multiple instructions need the same hardware resource.
- Pipeline depth (ideal speedup) is $\text{Speedup} = \text{Pipeline depth} / (1 + \text{pipeline stall cycles per instruction})$.
- Operand (result) forwarding/bypassing reduces data hazards without waiting for register write-back.
- Execute (EX) is branch conditions and target addresses are typically resolved in the execute stage.
- Deep pipelines improve clock rate (throughput) but increase hazard/branch (stall) penalties.
- Instruction-level ILP lets multiple instructions execute in parallel within a single pipeline.
- CPI is cycles per instruction, the average clock cycles each instruction needs.